# TARUN SINGH

**PhD Scholar | Department of Civil Engineering | IIT Roorkee**

Google Scholar | ORCID | LinkedIn | GitHub

## RESEARCH PROFILE

PhD researcher working on constitutive modelling of rate-dependent materials, with emphasis on finite-deformation frameworks, rheological connections, evolving natural configurations, and thermodynamically consistent modelling of viscoelastic solids.

## EDUCATION

**PhD, Civil Engineering | 2023–Present**
Indian Institute of Technology Roorkee
Research: Constitutive Modelling of Rate-Dependent Materials | Supervisor: Dr. Sandipan Paul

**M.Tech., Structural Engineering | 2020–2022**
National Institute of Technology Jalandhar | Supervisor: Dr. Navdeep Singh
Thesis: Carbonation Resistance of Concrete Made with Iron Slag and Recycled Concrete Aggregate Dr. Sandeep Chaudhary

**B.Tech., Civil Engineering | 2015–2019**
Rewa Engineering College, Rewa, Madhya Pradesh
Project: Development of Porous Asphalt Pavement

## RESEARCH EXPERIENCE

- PhD research on constitutive modelling of rate-dependent materials and finite-deformation viscoelasticity.
- Development of constitutive frameworks using rheological connections and multiple/evolving natural configurations.
- Formulation and numerical integration of nonlinear constitutive equations, with parameter identification using experimental response data.
- M.Tech. research on durability and performance of concrete incorporating iron slag and recycled concrete aggregates.

## SELECTED PUBLICATIONS

**Singh, T. and Paul, S. (2026).** "Constitutive modeling of Kelvin–Voigt type viscoelastic solids undergoing finite deformations using the theory of evolving natural configurations." *International Journal of Solids and Structures, 341, Article 114336.* DOI: https://doi.org/10.1016/j.ijsolstr.2026.114336

**Singh, T. and Paul, S. (2025).** "On the extension of the concept of rheological connections to a finite deformation framework using multiple natural configurations." *International Journal of Engineering Science, 212, Article 104267.* DOI: https://doi.org/10.1016/j.ijengsci.2025.104267

**Singh, N., Singh, A., Ankur, N., Kumar, P., Kumar, M., and Singh, T. (2022).** "Reviewing the properties of recycled concrete aggregates and iron slag in concrete." *Journal of Building Engineering, 60, 105150.*

**Singh, N., Ankur, N., Singh, T., Kumar, M., Singh, A., and Kumar, P. (2022).** "Non-destructive evaluation of iron slag concrete containing recycled concrete aggregates." *Indian Concrete Journal, 96(11).*

**Singh, N., Singh, T., Kumar, M., Singh, A., and Kumar, P. (2022).** "Investigating the fresh state performance of concrete containing iron slag and recycled concrete aggregates." *Materials Today: Proceedings, 65, 1467–1477.*

## CONFERENCE PRESENTATIONS

**Singh, T. and Paul, S. (2026).** "Thermodynamically Consistent Constitutive Model for Viscoelastic Solids Under Finite Deformation Using Evolving Natural Configurations." *7th Indian Conference on Applied Mechanics, IIT Kanpur.*

**Singh, T. and Paul, S. (2025).** "Challenges in extending rheological connections to a finite deformation theory." *14th International Symposium on Plasticity and Impact Mechanics (IMPLAST 2025), IIT Roorkee.*

## RESEARCH INTERESTS

Constitutive modelling of rate-dependent materials • Finite-deformation continuum mechanics • Finite-strain viscoelasticity and viscoplasticity • Rheological models and connections • Evolving and multiple natural configurations • Computational mechanics and finite-element implementation • Sustainable construction materials

## TECHNICAL SKILLS

**Scientific computing:** MATLAB, Python

**Finite-element and constitutive implementation:** Abaqus, UMAT, Fortran, FEniCSx

**Numerical methods:** Numerical integration of constitutive equations, parameter identification and model calibration